

CSE428 Lecture 13: Vision Transformer

1. **Why are images divided into patches in ViT?** What would happen if the entire image were treated as a single token?
2. Consider an image of size $224 \times 224 \times 3$ and a patch size of 16×16 .
 - a. How many patches will be generated?
 - b. What is the dimensionality of each flattened patch?
 - c. Including the **[CLS]** token, what is the total sequence length entering the Transformer encoder?
3. **What is the purpose of the [CLS] token in ViT?** How is it used for image classification?
4. Explain the role of the following components in a ViT encoder:
 - a. Multi-Head Self-Attention
 - b. Layer Normalization
 - c. MLP/Feed-Forward Network
 - d. Residual connections
5. Explain Multi-Head Self-Attention (MHSA). Why are multiple attention heads useful instead of using a single attention head?
6. Why is positional embedding required for each of the image patches? Explain.
7. **Compare CNNs and Vision Transformers.** Discuss at least five differences.
8. Why do CNNs have a strong **inductive bias** for images while ViTs have relatively weaker inductive biases?
9. A researcher has only **10,000 labeled medical images**. Would you recommend training a ViT from scratch? Explain your reasoning.
10. A ViT achieves **95% training accuracy but only 70% validation accuracy**. What problem is likely occurring? Give **three techniques** that could help address it.